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Reinforcement Learning for Channel Allocation: Digital Twin-Based Optimization in Real Large-Scale Wi-Fi Networks

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Abstract

Wi-Fi networks, which are based on the IEEE 802.11 standard, are among the most widely used wireless networks worldwide. In recent years, new standards have been introduced that offer a range of benefits to users. However, achieving full potential for these sophisticated standards hinges on the implementation of effective resource management strategies. This task is increasingly challenging due to the growing demand for high-performance services, particularly in massive networks, where the frequency channel allocation (CA) problem is of critical importance. Identifying the optimal channel configuration frequently represents a significant challenge. In this scenario, it is often the case that classical optimization approaches are often not sufficient. In such instances, machine learning (ML) has emerged as a promising solution. However, current solutions are insufficient for addressing real-time monitoring and bidirectional data and control flows, both of which are crucial for the management of the dynamic nature of modern networks. This study demonstrates that a reinforcement learning (RL)-based system, integrated with a digital twin (DT) methodology, can create a virtual representation of the network, allowing real-time adjustments without additional overhead. The MELODWIN system has been developed to optimize CA and enhance the performance of large-scale Wi-Fi networks. The validation results, obtained through simulations and in a real operational Wi-Fi network, provide clear evidence that MELODWIN improves performance in complex network environments. Compared to previous solutions, this approach integrates continuous monitoring and dynamic management capabilities, offering significant advantages. The proposed system establishes a foundation for future architectures that leverage RL and DT for wireless network management. With its ability to efficiently optimize performance in large-scale WiFi environments, MELODWIN holds promise for adaptation to other emerging wireless technologies.

Keywords

communication, networking and broadcast technologies, digital twin, machine learning, wi-fi, wireless

Reinforcement Learning for Channel Allocation: Digital Twin-Based Optimization in Real Large-Scale Wi-Fi Networks

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***Abstract*—Wi-Fi networks, which are based on the IEEE 802.11 standard, are among the most widely used wireless networks worldwide. In recent years, new standards have been introduced that offer a range of benefits to users. However, achieving full potential for these sophisticated standards hinges on the implementation of effective resource management strategies. This task is increasingly challenging due to the growing demand for high-performance services, particularly in massive networks, where the frequency channel allocation (CA) problem is of critical importance. Identifying the optimal channel configuration frequently represents a significant challenge. In this scenario, it is often the case that classical optimization approaches are often not sufficient. In such instances, machine learning (ML) has emerged as a promising solution. However, current solutions are insufficient for addressing real-time monitoring and bidirectional data and control flows, both of which are crucial for the management of the dynamic nature of modern networks. This study demonstrates that a reinforcement learning (RL)-based system, integrated with a digital twin (DT) methodology, can create a virtual representation of the network, allowing real-time adjustments without additional overhead. The MELODWIN system has been developed to optimize CA and enhance the performance of large-scale Wi-Fi networks. The validation results, obtained through simulations and in a real operational Wi-Fi network, provide clear evidence that MELODWIN improves performance in complex network environments. Compared to previous solutions, this approach integrates continuous monitoring and dynamic management capabilities, offering significant advantages. The proposed system establishes a foundation for future architectures that leverage RL and DT for wireless network management. With its ability to efficiently optimize performance in large-scale Wi-Fi environments, MELODWIN holds promise for adaptation to other emerging wireless technologies.**

***Index Terms*—Wi-Fi, digital twin (DT), reinforcement learning (RL), channel allocation.**

I. Introduction

Wi-Fi networks, which are based on the IEEE 802.11 standard, serve as the foundation for global wireless data communication. These networks are used by billions of users on a daily basis. Nowadays, a significant proportion of access points (APs) adopt the Wi-Fi 6 standard (IEEE 802.11ax), which represents the current state of the art for consumer devices. Wi-Fi APs are now ubiquitous in public spaces, residential areas, and offices in response to the growing demand for high-speed and low-latency communication. However, the rapid increase in traffic volume, coupled with the growing demand for improved capacity, latency, efficiency, and coverage, presents a significant challenge to the deployment and management of these networks [1]–[6].

In the past, Wi-Fi networks have been optimized using indicators at the network level, such as throughput and data rate. This approach has focused on the lower layers of the communications stack. However, the advent of contemporary applications, which are enhanced with georeferenced data, user preferences, and terminal measurements, necessitates the development of sophisticated decision-making processes, self-optimization, and fault detection mechanisms that integrate end-to-end (E2E) user-centric metrics. This paradigm shift generates large datasets that require the implementation of advanced analytical techniques to discern patterns and behaviors that are not feasible with traditional manual methods [7]–[9].

The growing complexity of Wi-Fi networks, compounded by uncoordinated deployments, network densification, and distributed management, introduces further performance challenges. These challenges require innovative approaches capable of addressing large-scale dynamic data environments. In this context, ML emerges as a promising solution, offering the potential to process raw data and optimize network operations where manual methods are inadequate [10], [11].

Congestion in the 2.4 GHz band has become so severe in many regions that this frequency is nearly unusable for high-rate data transfers [12]. This congestion is primarily due to the proliferation of devices and APs operating in the same spectrum, leading to increased interference and degraded performance. Furthermore, many users around the world experience frustration due to slow wireless networks [13] . The introduction of IEEE 802.11ax aims to address these issues by enhancing Wi-Fi performance in dense environments with numerous devices and overlapping networks. These insights underscore the challenges posed by the increasing number of Wi-Fi APs and stations (STAs), which complicate network management and contribute to a higher volume of complaints about Wi-Fi performance.

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CA represents a significant challenge in dense Wi-Fi environments, where a limited number of frequency channels must be shared among a large number of basic service sets (BSSs). Inadequate CA results in contention and interference among APs and STAs, thereby reducing overall throughput and the quality of the user experience. Despite the extensive exploration of centralized CA schemes, their practical applicability in dynamic, real-world scenarios remains constrained due to scalability and adaptability issues [14]–[18].

Real-time monitoring and management of Wi-Fi networks is a crucial aspect in addressing their inherently dynamic nature. However, current solutions often lack comprehensive mechanisms for real-time monitoring, bidirectional data flows, and adaptive control. The DT paradigm offers a promising framework for addressing these shortcomings. A DT represents a virtual replica of the physical network, enabling real-time monitoring, analysis, and optimization. Using continuous bidirectional data flows between physical and virtual networks, DTs enable efficient network reconfiguration without introducing additional overhead [19]–[22].

In addition, it is not only important to establish bidirectional flows between a DT and the real network, but it is also important to take into account the problem that the most up-to-date information is not always obtained from the network, since reports are often not obtained regularly or with good quality. This requires dynamic strategies based on the information received to optimize a live operating network.

This study focuses on large-scale Wi-Fi networks operating in the 2.4 GHz band, where inter-channel interference is considered a major challenge. The CA problem is addressed with the objectives of maximizing throughput and minimizing network congestion, particularly for outlier users. The inherent complexity and dynamic behavior of Wi-Fi networks are highlighted as motivations to explore RL. However, the major motivation here is to find optimization strategies in realistic network states, such as when the incoming information is not up-to-date or does not arrive in good enough condition to train an RL algorithm appropriately for the network. Therefore, the key is to obtain optimizations that are sensitive to the information obtained from the operational network, whatever it may be, since in many cases this is what leads to optimization errors, not only because the algorithm is based on outdated or poor quality information, but also because if it relies on a DT, it may not replicate the network with the appropriate conditions.

In order to address the aforementioned challenges, the implementation of an RL-based system, supported by a DT framework, is proposed for the dynamic and efficient allocation of channels in Wi-Fi networks. The principal contributions of this study are as follows:

- An automated CA optimization system that integrates an RL algorithm with a Network Digital Twin (NDT) is presented in a decision-theoretic framework. This system is unique in its incorporation of two different optimization strategies, depending on the type of information obtained from the network, making it very robust and integrable in all types of operational Wi-Fi networks. In addition, the integration process for use in any operational network is presented.
- Simulation results are shown demonstrating the potential of the proposed system to optimize large-scale operational networks against others showing different figures of merit that are considered key to the good performance of Wi-Fi networks.
- The system is then evaluated on a real-world live operational large-scale Wi-Fi network of 92 nodes, already optimized with current techniques. The results show relevant improvements on the main performance metrics of the network.

These corroborate the ability of the proposed system, named MELODWIN, to improve network service not only at the simulation level but also in real operational large-scale networks, validating its relevance.

The remainder of this paper is organized as follows: Section II summarizes the related work, while Section III presents the proposed optimization system in detail. Section IV shows the results of the simulation-based evaluation of the mechanisms. Section V validates the system in the live operational network. Finally, Section VI concludes the paper and discusses possible future directions.

II. Related Work

The growing complexity and density of Wi-Fi deployments have driven the development of numerous techniques for optimizing wireless networks. Among these, three key areas have attracted significant research attention: CA strategies tailored to Wi-Fi constraints, the use of RL to enable dynamic and adaptive control, and the application of DT technology to safely simulate and improve network behavior. This section provides a review of relevant work in each of these domains, highlighting existing methods, their limitations, and how they inform the proposed approach.

A. The Channel Allocation problem in Wi-Fi Networks

Channel allocation in Wi-Fi networks has been the subject of considerable attention in recent literature, with a number of algorithms proposed to enhance network performance. Connectivity management, including CA, is widely acknowledged as a complex task due to the interaction between link configuration and neighboring network performance [15].

Several studies have proposed potential solutions to address this challenge. In reference [23], a cloud-assisted dynamic CA system was proposed as a means to improve performance and user experience between Wi-Fi networks groups. The system is dependent upon the periodic acquisition of channel availability data and the collation of neighbor information by APs operating on 2.4 GHz and 5 GHz channels. Processing of these data sets on a centralized cloud platform enables the generation of channel state predictions and the determination of the optimal CA for all networks within the group. Similarly, an alternative method was described in reference [24], where throughput was used as an evaluation metric. In this approach, all channel configurations were scored and the configuration with the highest score was selected. However, these methods assume periodic and up-to-date measurement collection, which may not be feasible in dynamic operational networks. Furthermore, external APs interference or rapidly changing network conditions may introduce additional complexity to CA decision-making.

While manual or heuristic approaches can partially address these challenges, the integration of large-scale heterogeneous data sources necessitates the use of machine learning (ML) techniques to extract meaningful patterns and trends [25]. However, most existing works do not consider the impact of data staleness or delayed information reporting, which is common in real Wi-Fi deployments and can significantly affect CA effectiveness.

B. Reinforcement Learning to optimize Wireless Network

RL has been shown to have considerable potential for optimizing wireless networks, due to its capacity to learn dynamic policies in evolving environments. ML, as a more comprehensive technique, has been widely adopted in wireless communication, driven by its success in solving complex tasks without the need for explicit programming [26], [27].

RL-based solutions offer several advantages. These include the ability to extract spatial and sequential features from metrics such as the received signal strength indicator (RSSI) [28], the ability to approximate traditional high-complexity algorithms with lower complexity [29], and the potential for self-organizing networks through multi-agent RL. Additionally, RL can achieve fast network control based on learned policies [30]. Transfer learning further improves RL-based approaches by accelerating the learning process in new tasks through the reuse of prior knowledge, thus addressing spatial and temporal correlations common in wireless systems [27].

López-Raventós and Bellalta [31] investigated multiarmed bandit (MAB)-based decentralized CA and AP selection methods in enterprise WLANs. Their approach used Thompson sampling algorithms for AP-side decision-making. Another work, presented in [32], utilized deep RL (DRL) for CA in dense multi-BSS WLAN deployments, using a central controller to maximize throughput. Although these studies demonstrated promising results in simulations, real-world deployment faces challenges such as outdated training data and the highly dynamic nature of operational networks.

In [33], resource allocation in WLANs has been studied and performance improvement has been achieved. An algorithm has been proposed for WLANs that couples power control and RL-based channel allocation to improve performance. The algorithm ds validated in a scenario of 15 APs in a $100\,\text{m} \times 100\,\text{m}$ environment and performing the actions directly on the network. In [34] a practical joint CA and TPC scheme is proposed that achieves good performance in terms of packet transmission delay and energy consumption, in WLAN environments with 13 APs where controlled APs and uncontrolled APs coexist. The proposed scheme adopts a hybrid approach of centralized training and distributed execution, where the DRL model is trained by a central trainer and converted into a lightweight form factor for APs with limited capacity. Each controlled AP makes its own AR decision using the lightweight DRL model distributed by the trainer. The state and action spaces of the DRL model are designed to be unaffected by the number of APs. The method is validated by simulation, and actions are performed directly on the network.

The validation of RL algorithms in live operational large-scale Wi-Fi networks scenarios is key because although simulations are the first step in the evaluation phase it has to be taken to a real and operating environment; in addition to that these algorithms dump actions on the operational network when that can completely degrade the performance until it finds the most appropriate action.

C. Digital Twins to support Network Optimization

Digital Twin technology has emerged as a transformative tool in sectors such as healthcare, aviation, and telecommunications [35]. In the telecommunications domain, DT has been applied to 5G [36] and Wi-Fi networks [37], facilitating enhanced network design and maintenance through virtual replication of physical systems.

In wireless networks, the reliance on ML-based solutions without DT introduces potential risks, such as performance degradation due to real-time training in operational systems. In contrast, DT enables predictive analysis and simulation-based optimization without affecting live networks [38].

Several studies have used a combination of DT and RL for the purpose of optimizing wireless networks. For example, a DT-based framework for mobile networks was proposed in [39]. In this framework, DT simulations were employed to evaluate optimization decisions generated by RL and expert systems, enabling higher rewards while avoiding adverse effects on live network performance. Another study, described in [40], introduced a DT-enabled Wi-Fi network (DTWN) architecture that incorporates Q-learning for transmit power control (TPC). This architecture achieved real-time monitoring and optimization, demonstrating improvements in total interference and throughput as the number of users increased.

The integration of DT with RL offers a promising approach to wireless network optimization, which addresses different challenges. DT architectures with RL have been studied preliminarily; however, no validations have been performed on massive networks where a degree of difficulty is added to the optimization. In addition, the age of the information collected from the network has not been taken into account, since in many occasions in live operational networks the reports obtained are not the most recent. Also, the type of optimization, the time at which the network is configured, and the time at which data is collected from the network must be taken into account.

III. System Overview

The MELODWIN system is designed to enable intelligent and scalable CA in large-scale Wi-Fi networks through the integration of DTs and RL. As shown in Fig. 1, the architecture is structured around two main components: (1) the Twin Decision and RL engine, responsible for decision-making and learning, and (2) the NDT execution environment, where simulated evaluations are performed based on real-world inputs.

The Twin Decision module is responsible for receiving real-time reports and performance metrics from the Wi-Fi network, with the objective of optimizing its performance. The incoming data is first analyzed to estimate its freshness (i.e. data age), which is crucial for determining the reliability of the simulations. Following a thorough analysis, MELODWIN determines the most suitable DT execution strategy. Furthermore, it is responsible for the configuration of the RL algorithm. This process encompasses the definition of reward functions and action strategies for the learning agents.

Subsequent to the definition of the strategy, the selected data is forwarded to the NDT layer. Depending on the state of the network and the availability of fresh data, MELODWIN selects between two modes: an Alien-Free DT, constructed using recently updated network data, or multiple DT instances, generated using older data and different assumptions about the environment. The term 'alien' is used to describe APs that are external to the user's control, i.e. whose configuration cannot be modified or observed.

A. Learning-based Strategy in MELODWIN

RL has been selected for the development of this system because of its well-established suitability to address challenges related to the optimization and management of wireless networks. The highly dynamic nature of Wi-Fi networks, influenced by factors such as device mobility, interference, and congestion, makes RL an effective tool for adaptation and continuous learning. RL algorithms are capable of adjusting to changes in the environment and autonomously optimizing network performance based on specific metrics. This adaptability is of paramount importance for maintaining optimal performance in diverse network environments, with the capacity to accommodate fluctuations in traffic load or Quality of Service (QoS) requirements.

The Q-learning approach to RL is particularly effective due to its scalability and computational efficiency, which are essential for real-time adjustments in dynamic environments. The simplicity of Q-Learning makes it a computationally feasible approach while still effectively handling stochastic environments. Within Q-learning, the epsilon-greedy policy is used to balance the exploration and exploitation of actions. This policy ensures constant exploration of new actions without requiring complex probability distribution calculations. The epsilon value can be adjusted to manage the exploration-exploitation balance, allowing flexibility for different scenarios and system preferences.

A principal element of RL systems is the environment model, which simulates the behavior of the network. Based on a state and action, the model predicts the subsequent state and the corresponding reward. This model-based approach contrasts with model-free methods, where the agent learns through trial and error. The model-based approach enables proactive planning by allowing the agent to predict future outcomes before taking action, thus optimizing performance.

The integration of a DT into the optimization system is critical due to the intricate and evolving nature of wireless networks. The DT furnishes an accurate and real-time virtual representation of the network, thereby facilitating rapid adjustments. Moreover, it reduces the need for intrusive modifications to the actual network by enabling experimentation and modifications in a virtual environment prior to implementation. This facilitates real-time adaptability and accelerates the learning process, leading to better convergence of the optimization system. The combination of ML and DTs thus enhances operational efficiency and allows for more agile and accurate handling of the dynamic challenges inherent in wireless networks.

The Q-learning algorithm employs several parameters in its operational processes. On the one hand, there are the agents that perform actions within the environment. The environment then returns a series of rewards, which are stored in a table called the Q-Table. This table contains information about the states of the environment and the possible actions that can be performed within those states. In this case, the agent is the AP that performs the channel selection action and the states are the conditions of the different APs within the selected cluster that are to be optimized. Therefore, the states (s_t) are:

$$\mathrm{s}_t = \{AP1...APN\} \tag{1}$$

The actions (a_t) are as follows:

$$\mathrm{a}_t = \{channel1...channel11\} \tag{2}$$

Consequently, Q-Table is constituted with these states (s_t) and actions (a_t).

The action a_t taken at time t in state s_t is determined by:

$$\mathrm{a}_t \text{ in } \mathrm{s}_t = \begin{cases} \text{argmax}_{a_t} Q_t(s_t, a_t) & \text{with probability } 1 - \epsilon_t \\ \text{random action } (s_t, a_t) & \text{with probability } \epsilon_t \end{cases} \tag{3}$$

In this context, the initial values of ϵ_t are defined as follows:

$$\epsilon_{\text{decay}} = 0.98 \tag{4}$$

$$\epsilon_{\text{initial}} = 1 \tag{5}$$

$$\epsilon_t = \epsilon_{\text{initial}} \cdot \epsilon_{\text{decay}} \tag{6}$$

Once the AP has selected a given channel, the network with that new state is simulated in the DT. The simulator returns the figure of merit for the throughput and airtime of each of the AP. With these normalized values, it obtains the reward, $\text{global}_{\text{reward}}$ defined as:

$$\bar{Th}_{\text{tot}} = \frac{Th_{\text{AP}_{\text{mean}}} - Th_{\text{min}}}{Th_{\text{max}} - Th_{\text{min}}}, \quad 0 < \bar{Th}_{\text{tot}} < 1 \tag{7}$$

$$\bar{Air}_{\text{tot}} = \frac{Air_{\text{AP}_{\text{mean}}} - Air_{\text{min}}}{Air_{\text{max}} - Air_{\text{min}}}, \quad 0 < \bar{Air}_{\text{tot}} < 1 \tag{8}$$

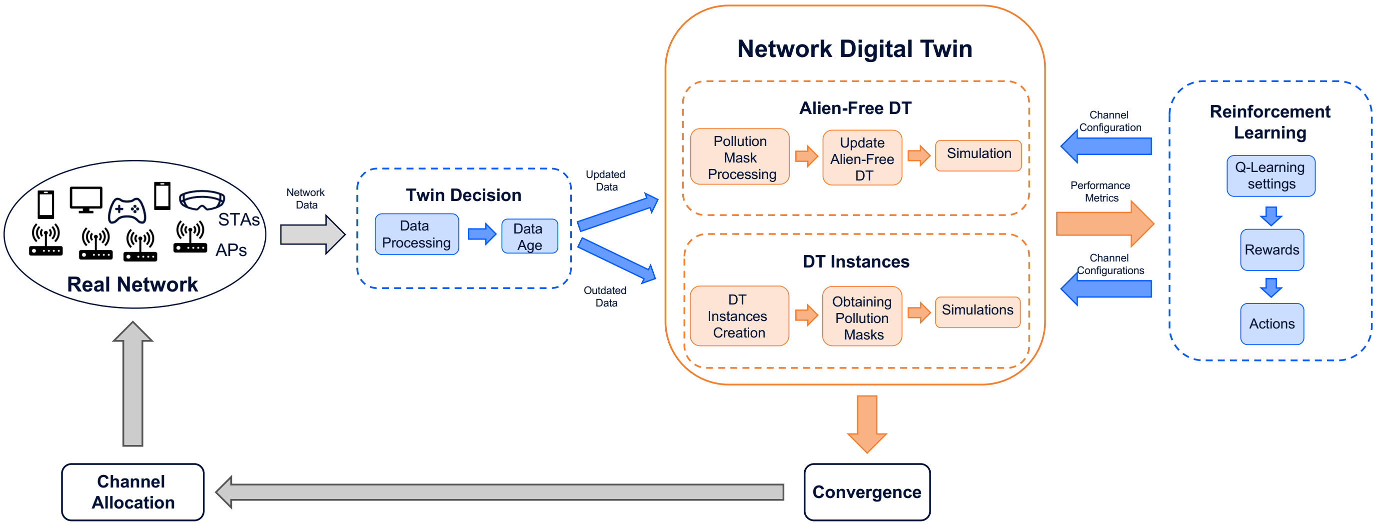


Fig. 1: An overview of the MELODWIN system.

$$\text{global}_{\text{reward}} = v_{1_{\text{tot}}} \cdot \bar{Th}_{\text{tot}} - v_{2_{\text{tot}}} \cdot \bar{Air}_{\text{tot}} \tag{9}$$

$$0 < \text{global}_{\text{reward}} < 1 \tag{10}$$

Where $\bar{Th}_{tot}$ represents the normalized total throughput reward, which measures the overall network performance, $Th_{\text{AP}_{\text{mean}}}$ is the average network throughput per AP, $\bar{Air}_{\text{tot}}$ is the normalized total airtime reward, and $Air_{\text{AP}_{\text{mean}}}$ is the average airtime per AP.

In addition, the individual reward per AP is also considered, by first obtaining a reference value of airtime and throughput of the network with the initial configuration (before optimizing). APs that are below this throughput reference value and those above this airtime reference value are counted for penalty to obtain the $\text{indiv}_{\text{reward}}$,

$$\bar{Th}_{\text{ind}} = \frac{NTh_{\text{out}} - Th_{\text{ind}_{\text{min}}}}{Th_{\text{ind}_{\text{max}}} - Th_{\text{ind}_{\text{min}}}}, \quad 0 < \bar{Th}_{\text{ind}} < 1 \tag{11}$$

$$\bar{Air}_{\text{ind}} = \frac{NAir_{\text{out}} - Air_{\text{ind}_{\text{min}}}}{Air_{\text{ind}_{\text{max}}} - Air_{\text{ind}_{\text{min}}}}, \quad 0 < \bar{Air}_{\text{ind}} < 1 \tag{12}$$

$$\text{indiv}_{\text{reward}} = v_{1_{\text{ind}}} \cdot \bar{Th}_{\text{ind}} - v_{2_{\text{ind}}} \cdot \bar{Air}_{\text{ind}} \tag{13}$$

$$0 < \text{indiv}_{\text{reward}} < 1 \tag{14}$$

In this context, $\bar{Th}_{\text{ind}}$ represents the normalized throughput individual reward, NTh_{out} signifies the number of APs that fall outside the throughput range, $\bar{Air}_{\text{ind}}$ denotes the normalized airtime individual reward, and $NAir_{\text{out}}$ indicates the number of APs that are not within the airtime range.

Furthermore, alien interference is considered in each AP, which is obtained from the alien pollution mask channel of the manageable network. In this case, the reward is maximized when the difference between the received signal strength indicator (RSSI) of the AP from its STAs and the pollution received from the aliens on a given channel exceeds 10 decibels (dB). Therefore, the $\text{aliens}_{\text{reward}}$ is calculated as follows:

$$\bar{A} = \frac{(\text{RSSI}_{\text{AP}} - \text{RSSI}_{\text{alien}}) - \text{RSSI}_{\text{min}}}{\text{RSSI}_{\text{max}} - \text{RSSI}_{\text{min}}} \tag{15}$$

$$\bar{A} = \begin{cases} 1 & \text{if } (\text{RSSI}_{\text{AP}} - \text{RSSI}_{\text{alien}}) > 10dB \\ \bar{A} & \text{if } (\text{RSSI}_{\text{AP}} - \text{RSSI}_{\text{alien}}) \leq 10dB \end{cases} \tag{16}$$

The reward for aliens is given by:

$$\text{aliens}_{\text{reward}} = \bar{A} \tag{17}$$

$$0 < \text{aliens}_{\text{reward}} < 1 \tag{18}$$

Where $\text{RSSI}_{\text{alien}}$ denotes the average RSSI of the alien per channel, RSSI_{AP} refers to the average RSSI of the STAs in the AP, RSSI_{min} represents the minimum RSSI difference between the AP RSSI and the alien RSSI, and RSSI_{max} represents the maximum RSSI difference between the AP RSSI and the alien RSSI.

Finally, a reward is added (w_4) for key channels 1, 6 and 11 since these are the only channels in the 2.4 GHz spectrum that have no frequency overlap. With this reward, APs can be incentivized to have a certain priority for these key channels.

All rewards described above form the total reward:

$$\begin{aligned} r(s_t, a_t) = {} & w_1 \cdot \text{global}_{\text{reward}} \\ & + w_2 \cdot \text{indiv}_{\text{reward}} \\ & + w_3 \cdot \text{aliens}_{\text{reward}} \\ & + w_4 \end{aligned} \tag{19}$$

Where the weights are normalized as:

$$w_1 + w_2 + w_3 + w_4 = 1 \tag{20}$$

The definition of rewards within the optimization algorithm allows the pursuit of disparate approaches to optimization. Three distinct approaches to optimization are defined, each employing a different set of channels. In the first approach, channels 1, 6, and 11 are exclusively utilized. In the second approach, these channels are given a certain preference, though they are not the sole determining factor. In the third approach, any channel can be employed without restriction. The target APs channel range optimization approaches are defined by modifying the reward weights as follows:

$$\textbf{1,6,11} = \begin{cases} w_1 = 0.25 \\ w_2 = 0.25 \\ w_3 = 0.25 \\ w_4 = 0.25 \end{cases} \tag{21}$$

$$\textbf{1,6,11 pref.} = \begin{cases} w_1 = 0.33 \\ w_2 = 0.33 \\ w_3 = 0.33 \\ w_4 = 0.01 \end{cases} \tag{22}$$

$$\textbf{1 to 11} = \begin{cases} w_1 = 0.33 \\ w_2 = 0.33 \\ w_3 = 0.33 \\ w_4 = 0.33 \end{cases} \tag{23}$$

The total reward, in conjunction with the learning rate ($\alpha = 0.1$) and the discount factor ($\gamma = 0.9$) whose values have been obtained through empirical investigation, constitutes the reward function:

$$Q(s,a) = (1-\alpha)\cdot Q(s,a) + \alpha \cdot \left(r(s_t, a_t) + \gamma \cdot \max_a Q(s', a) \right) \tag{24}$$

Let a_t be the action taken in state s_t and r_t be the reward received in state s_t, the algorithm converges if it is satisfied:

$$a_{t-4} = a_{t-3} = a_{t-2} = a_{t-1} = a_t \tag{25}$$

$$r_{t-4} = r_{t-3} = r_{t-2} = r_{t-1} = r_t \tag{26}$$

The complete proposed Q-learning process is shown in Algorithm 1 where all the steps to find convergence in terms of channel assignment optimization are present.

B. Network Digital Twin Approach

The distinctive reliance on ML in wireless network environments without the assistance of a DT poses considerable challenges. Direct testing in the operational network can result in performance degradation during critical periods, as real-time adjustments and training can impede the quality of service.

As discussed and illustrated in Fig. 1, the RL algorithm interacts with the NDT, which is composed of two DT strategies. Reality and DT age differently, leading to divergence. The real network evolves over time, which is why different strategies are needed depending on the age of the real network

Algorithm 1 Q-Learning epsilon-greedy Model-Based Algorithm

```
Initialize ϵ = 1 and Q-Table values to −1
Establish ϵ_min = 0.01, ϵ_decay = 0.98, α = 0.01 and γ = 0.9.
Identify the APs of the cluster (APs_cluster) to be optimized
Establish values of weigths: w1, w2, w3 and w4.
repeat
    Use Digital Twin with the initial configuration and models to obtain actual figures of merit and thresholds: Th_APmean, Air_APmean, NTh_out and NAir_out.
    for AP to APs_cluster do
        repeat
            if rand() > ϵ then
                Select the action a_t = arg max_a Q(s_t, a).
            else
                Randomly select an action a_t ∈ A.
            end if
            Use Digital Twin with action select, models and reports to obtain figures of merit and thresholds: Th_APmean, Air_APmean, NTh_out and NAir_out.
            Obtain all rewards global_reward, indiv_reward, alien_reward and key_channels and apply the weights.
            Introduce the reward R from AP s_t and channel a_t in the reward function to obtain the value Q(s_t, a_t) from the Q-Table.
            if ϵ >= epsilon_min then
                Epsilon reduces, less randomness
            end if
        until convergence
    end for
until end all clusters
```

information. This can be observed in Fig. 2, where the impermanence of APs that are not available every day is shown with respect to the proportion of APs that are not available on a given day. If the data is up-to-date, the DT strategy without alien is chosen, and if it is not up-to-date, the DT Instances strategy is selected.

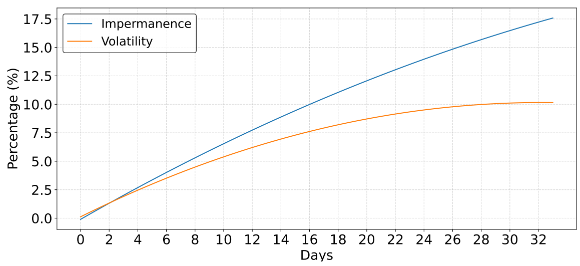


Fig. 2: Uncertainty cone showing the volatility and impermanence of the network over the days.

If the Alien-Free DT strategy is chosen, the updated alien pollution mask (see Fig. 3) provided by the real network is used to know the pollution added by these APs to the frequency channels and thus not having to model the position of the aliens. On the other hand, if the data is not updated, the Alien-Free DT is not sufficient, since the alien mask will be outdated and will not correspond to the pollution of the real network. Therefore, the system will choose the DT Instances strategy, where it will perform N simulations to reliably obtain the channel assignment that best suits the network.

Legend:
Available Channel
Problematic Channel

2.4 GHz Channels

AP ID	1	2	3	4	5	6	7	8	9	10	11
AP1	-52	-52	-62	-79	-89	-89	-86	-75	-58	-48	-48
AP2	-44	-44	-54	-62	-54	-55	-54	-59	-60	-57	-50
AP3	-64	-61	-62	-62	-72	-87	-77	-67	-64	-57	-57
AP4	-54	-54	-62	-66	-67	-67	-67	-77	-94	-103	-106
AP5	-59	-59	-69.	-71	-61	-61	-61	-71	-88	-93.	-77
AP6	-92	-87	-76	-60	-50	-50	-50	-60	-64	-55	-55
AP7	-57	-56	-59	-58	-52	-53	-53	-56	-54	-47	-44
AP8	-88	-83	-81	-75	-71	-60	-43	-33	-33	-33	-43
AP9	-64	-64	-74	-91	-102	-104	-103	-92	-75	-65	-65
AP10	-66	-66	-76	-72	-62	-62	-62	-69	-55	-45	-45

Fig. 3: Example of channel alien mask pollution.

IV. Simulation Evaluation

Before obtaining results from a real operational network, it is imperative to validate the system through a synthetic scenario. This allows for the evaluation of the feasibility and performance of the proposed solution in a controlled environment.

A. Simulation Settings and Scenario

The simulation configuration parameters are defined in Table I and have been selected to obtain the results of this subsection.

Two simulation durations are considered: an initial run of 48 hours for preliminary comparison between methods, followed by an extended simulation of 128 hours. This longer duration allows for observing how network behavior evolves over time and how the system adapts to dynamic conditions. The simulations are based on the IEEE 802.11ax standard and focus on channel selection in the 2.4 GHz band, with a transmit power of 20 dBm—a typical real-world scenario, and one prone to high interference.

The simulator operates using an event-driven engine: performance reports from a randomly selected AP in the network are generated every 5 minutes, and the traffic demand is updated every hour. The optimization event is triggered daily at 6:00 a.m., reflecting typical operational practices, as this is when the network is usually least active. Residential IPTV traffic is modeled, which reflects the most common usage pattern in the network. Additionally, wall attenuation and wall length are included as environmental parameters to account for signal propagation effects.

A synthetic scenario comprising 12 APs and 12 aliens with 3 associated STAs (see Fig. 4) has been constructed although

TABLE I: Simulation configuration parameters.

Parameter	Value
Simulation Time	48, 128 h
Band	2.4 GHz
Standard	802.11ax
Transmit power	20 dBm
Event time value traffic change	3600 s
Event time value send report	300 s
Optimization time	6:00 a.m.
Traffic profile	IPTV residential
Outer wall length	9 m
Outer wall attenuation	10 dB
Inner wall length	4 m
Inner wall attenuation	5 dB

for preliminary comparison the aliens are not added to the simulation, for the purpose of observing the behavior of the optimization approach, comparing it with manual optimization approaches and assessing its potential to improve CA in such scenarios.

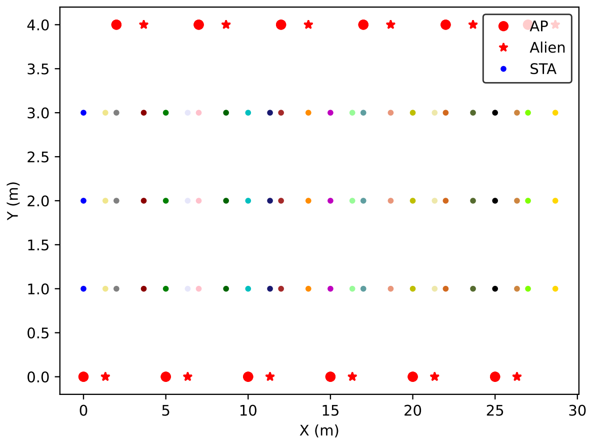


Fig. 4: Synthetic scenario generated to validate the optimization approaches and composed of 12 APs, 12 aliens and 3 STAs per AP.

B. Comparison with existing Strategies

Before conducting the simulations for different optimization approaches, a preliminary comparative analysis was performed to evaluate the benefits of incorporating a DT alongside Q-learning into the optimization architecture. This approach was compared with existing strategies, including standalone Q-learning, expert knowledge, random channel selection, and unboxing (where all APs operate on the same channel, representing the worst-case scenario).

These preliminary simulations were run over a 48-hour period in the scenario of Fig. 4 (without aliens), with a demanding traffic load designed to evaluate which approach optimizes performance most effectively under these conditions. The evaluation metrics were based on two key thresholds:

- The percentage of hours in which the network experienced congestion, defined by a realized airtime threshold of 0.95 seconds. If an AP needs more than 0.95 seconds of airtime it could be considered congested.
- The percentage of hours in which the network failed to achieve a throughput of at least 100 Mbps overall.

Fig. 6 shows how MELODWIN is only 13% above the realized airtime threshold and the closest approaches are twice as long above that threshold and therefore the network is congested longer. In terms of throughput using MELODWIN the network is only 35% of the time below 100 Mbps compared to 44% of the expert knowledge approach, and this approach is unsustainable in large-scale networks as it is unfeasible to optimize and manage networks with a large amount of information manually.

With respect to the other approaches, unboxing was expected to be the one with the worst performance, random channels not having a large number of APs does not have a bad performance compared to other approaches as there may be combinations in which there are not too many overlaps and Q-learning having to perform the channel allocation process on the real network, that lowers the performance and does not converge in the best channel allocation solution.

The results of these preliminary simulations showed that the integration of a DT with Q-learning has the potential to outperform the other approaches in terms of performance in more demanding networks, which justifies its selection for further evaluation in subsequent simulations with different optimization approaches and in more realistic scenarios.

C. MELODWIN Evaluation in Simulation

Having seen that MELODWIN may have a higher potential than other current optimization techniques, it is going to be evaluated in a more exhaustive way so that the performance can be observed over a longer time with a network that is a priori optimized and with aliens adding interference.

For the performance results, an optimal CA initially for the controllable APs and different choices of CA for the aliens have been taken into account.

The optimal CA pattern that would be obtained manually for APs situated in Fig. 4 at y=0 is [1, 6, 11, 1, 1, 6, 11], whereas for APs located at y=4, it is [11, 1, 6, 11, 6, 6, 1]. The issue arises when the CA pattern is determined a priori, as it can be solved with minimal effort. However, when unidentified alien channels emerge, affecting the performance of the network, the problem becomes more complex. Additionally, in scenarios with fewer APs, the CA pattern is relatively straightforward. However, as previously discussed, in denser networks, the problem becomes more intricate.

It has been taken into account when all the aliens are in the same channel (aliens one channel), when the aliens in y=0, when the aliens are optimized by means of the pattern [1,6,11] (aliens key channels) and when all range channels are selected randomly [1,2,...,10,11].

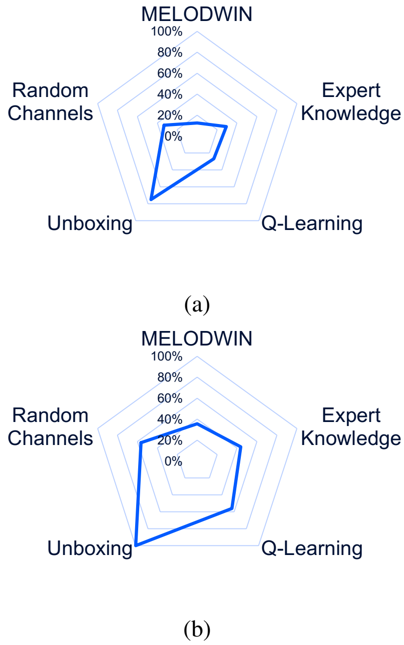


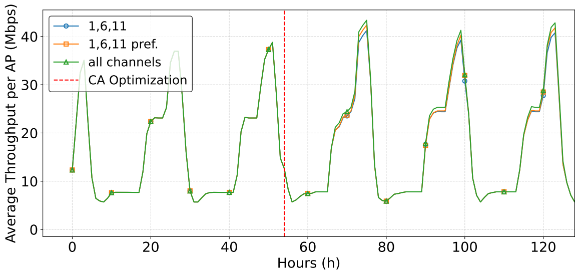
Fig. 5: Performance comparison with existing optimization strategies: (a) Comparison of the time that the average realized airtime of the network is above 0.95 seconds; (b) Comparison of the time that the average network throughput is below 100 Mbps.

In this case, the results focus more on the average of the individual APs and not so much on the network as a whole, since the objective is to be able to improve those APs that are not within the appropriate performance ranges. For a better understanding of the optimization potential, the throughput and congestion that defines how much extra airtime the AP needs to process the data is depicted. The comparison between the three optimization approaches defined above is depicted to further gain insight into which approach may work best.

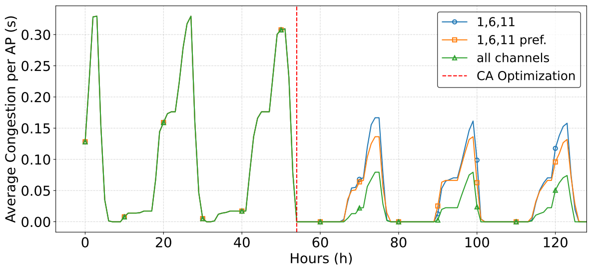
To obtain the results in Fig. 6 the scenario of Fig. 4 was simulated with the settings of Table I when the aliens are in the same channel. In terms of performance, it can be seen from the Fig. 6a that all three approaches result in an improvement of the initial state of the APs. In particular, it can be observed (Fig. 6b) that the ‘all channels’ approach has the effect of reducing congestion from approximately 30% to 7%.

The channels of the aliens were changed to the pattern [1,6,11] resulting in the generation of Fig. 7, which illustrates the performance in this particular instance. Fig. 7a illustrates that the throughput is enhanced in all optimization approaches. As illustrated in Fig. 7b, the ‘1,6,1 pref.’ and ‘all channels’ approaches result in a marked improvement in the initial allocation as it is reduced from approximately 11% to 4%. In the case of ‘1,6,11’, the initial congestion is reduced from only 11% to approximately 8%.

When the alien pollution is expanded by assigning random channels, the optimization process becomes more complicated (Fig. 8), as more channels with interference are introduced. An

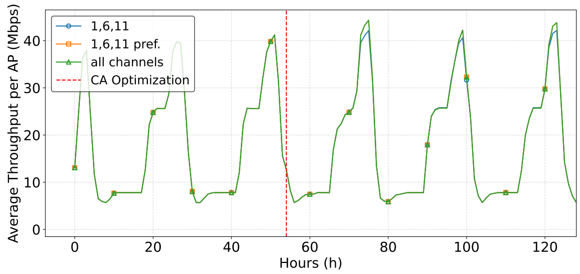


(a)

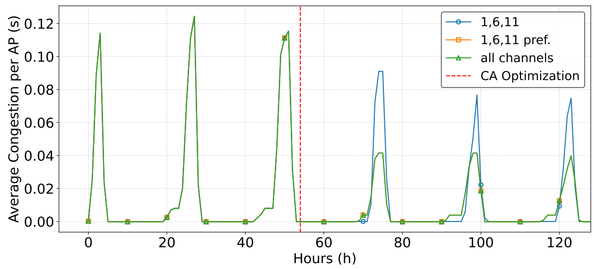


(b)

Fig. 6: Performance when all the aliens are in the same channel over time: (a) Average throughput per AP; (b) Average congestion per AP.



(a)



(b)

Fig. 7: Average performance when all the aliens are in key channels: (a) Average throughput per AP; (b) Average congestion per AP.

improvement in the initial allocation is observed with respect to performance across all three optimization approaches (see Fig. 8a). As for airtime congestion, it is observed in Fig. 8b that not only is it reduced from around 18% to a range between 8% and 12%, depending on the optimization approach, but the dwell time at high airtime values is also lower, making the graph narrower in all three optimization approaches, which improves the user experience.

The potential of MELODWIN to be able to improve the performance of Wi-Fi networks in scenarios where aliens add additional complexity due to the interference they introduce has also been tested. These results demonstrate that MELODWIN with different optimization approaches has the potential to enable APs to achieve possibly more demanding service requirements, since in the simulations performed with the described approaches the APs on average were able to spend more time with high throughput and even exceed 40 Mbps, in addition to improving airtime congestion reaching in some cases below 5% depending on the channels occupied by the aliens and the selected approach. All this suggests that MELODWIN may be ready to optimize a large-scale Wi-Fi network operating in real time.

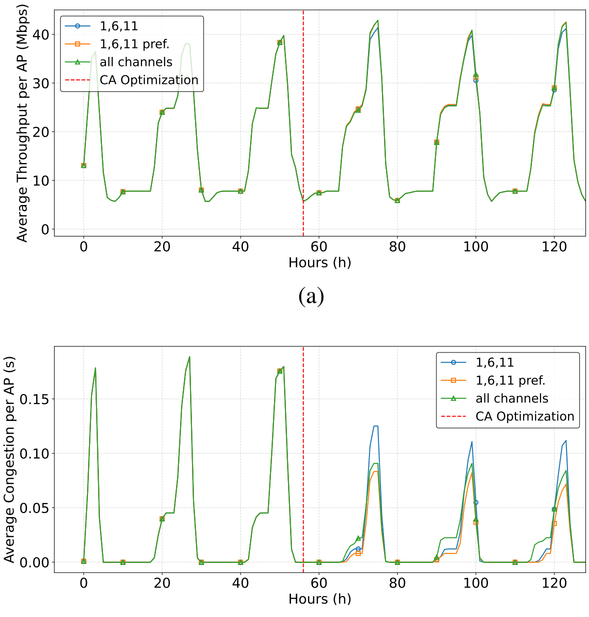
V. Validation in Live Operational Network

To assess the real-world applicability of MELODWIN, a comprehensive proof-of-concept (PoC) was conducted on a large-scale live operational Wi-Fi network already optimized with current techniques. This section describes the validation strategy and experimental conditions, outlines the process for integrating MELODWIN into an operational environment, and presents quantitative results that demonstrate the effectiveness of the proposed approach. The validation aims not only to test the performance of the MELODWIN system under realistic constraints, but also to evaluate its ability to handle partial or outdated data and adapt through the use of multiple DT instances.

A. *Strategy and Conditions*

It has been demonstrated that MELODWIN has the potential to optimize a live operational network, so a strategy is established to apply the system correctly in a PoC and to obtain a complete validation in large-scale Wi-Fi networks.

To perform the evaluation on a real operational network, the steps in Fig. 9 are followed. A cluster is selected as a candidate for optimization due to its congestion conditions. The next step is to observe and monitor it for a week to obtain the most up-to-date network data possible to achieve the best possible optimization. Based on the age of the network data, the appropriate strategy for optimization is selected, as seen in Fig. 1. With the network data, the offline phase of the DT optimization begins, during which the process of obtaining the best channel configuration in the 2.4 GHz band for the network is carried out. Once the list of configuration changes has been obtained, the relevant procedures are performed to update the network and add the new off-peak configuration. After the new configuration has been added to the network, the cluster is monitored again for a week to check if any changes have occurred in the network and to draw conclusions from the data obtained.



(a)

(b)

Fig. 8: Average performance when all the aliens are in random channels: (a) Average throughput per AP; (b) Average congestion per AP.

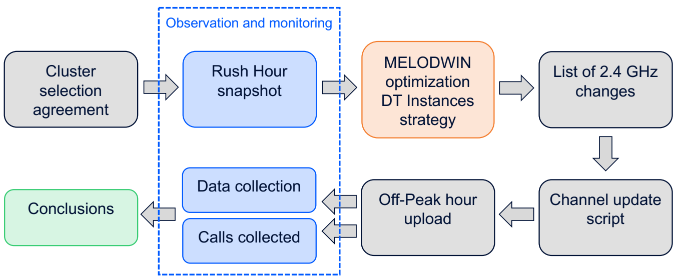


Fig. 9: Strategy followed in the PoC for using MELODWIN in a real operational Wi-Fi network.

The foundations of Wi-Fi optimization rely on improvements in user experience. A primary method to verify that the main goal is achieved could be the reduction in calls to the Communications Service Provider (CSP) call center. During the period of initial observation (pre-week monitoring), only one call was recorded, which was related to a fiber optic disconnection and not to Wi-Fi performance. During the execution and monitoring week, no calls were received from this cluster. As a result, the metrics could not be based on these indicators, but instead on how close the set of users was to underperforming conditions.

The best method to detect risks of underperformance was determined to be the analysis of the number of outliers in the cluster. Several ways to define outliers exist. An outlier can be defined as an element that falls far from or even outside the statistical limits. For the purposes of these experiments, the analysis started by counting how many elements fall outside $\mu + 3\sigma$ (99.7% of the cases). If no outliers were detected, the threshold was adjusted to $\mu + 2\sigma$ (95% of the cases). Three candidate parameters were considered for assessment: noise floor reported, airtime used by the APs on the channel, and total airtime used on the channel. Only the first two parameters were considered since the last two were deemed linearly related. Fig. 10 shows an example of noise outliers recorded on one day. The final decision was made to consider consistent $\mu + 2\sigma$ outliers for airtime and $\mu + 3\sigma$ for noise level. The difference between the two parameters could arise from the logarithmic nature of the second. The numbers were analyzed on a weekday basis to decouple from periodic human behavior.

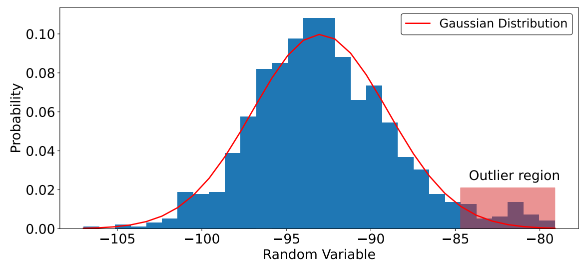


Fig. 10: Example of outlier region.

The strategy to be followed to apply MELODWIN to a live operational network and the associated conditions to validate the optimization system have been described in detail.

B. *Network Optimization Process*

Once the strategy and validation conditions have been described, the steps of the optimization process are presented.

As a first step in the process, the large-scale congested Wi-Fi network to be optimized in this PoC is selected. The real live operational network that has been selected for optimization is the one depicted in Fig. 11. It comprises 92 APs with an average of 3 STAs per AP, and a total of 231 aliens. The latter figure has been assumed to be representative of the actual number of STAs associated with each Alien, given that the actual data is not available.

In the network observation and monitoring step to extract data from the real network, if the data were recent and fully up-to-date, an alien-free DT strategy could be followed by incorporating the alien contamination mask from the real network as a reward in MELODWIN and simulating the alien-free scenario. This would result in the network performance shown in Fig. 12, using a 1,6,11’ optimization approach. A ‘1,6,11’ approach is used to validate MELODWIN in all approaches, since a priori it is the one with the worst results previously obtained although it also improves performance and may be the approach that gives more confidence to the operator as it is the most conservative. It should be noted that the network is initially optimized by the network operator and MELODWIN tries to improve on that previous optimization. As shown in Fig. 12a and 12b, the network not only improves in terms of throughput and airtime, but Fig. 12c illustrates

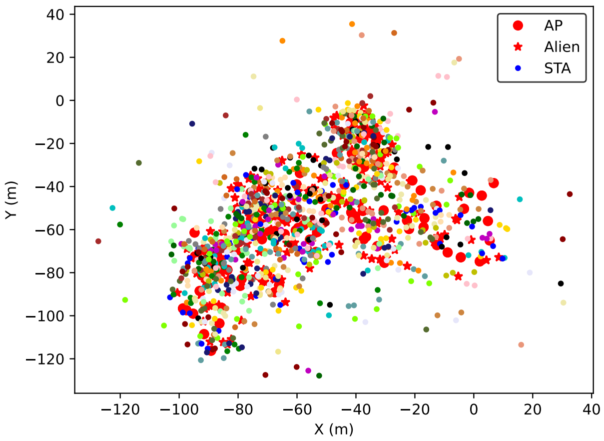


Fig. 11: Real live operational large-scale Wi-Fi network with AP positions and recreation of aliens and STAs positions.

the reduction of congestion. It shows the number of APs that are outliers in congestion, i.e., they are outliers in atypical performance and therefore may not be providing good service. It can be seen that the number of outlier APs in terms of congestion decreases from 4 to zero.

However, in this case, network data were obtained that were not completely up-to-date and not completely recent, therefore, a strategy of creating multiple DTs (DT Instances) was adopted, modeling also the aliens. For the reasons already explained, the optimization approach of using only channels 1, 6 and 11 for channel assignment was chosen. In addition, MELODWIN incorporates a minimum distance method to select the simulation combination that most closely matches most channel selections.

In the context of the DT instance strategy, 25 DTs have been constructed based on real network data, on which simulations have been performed to see how the network behaves. Fig. 13 illustrates the performance of the selected DTs. It reveals that, in some cases, the performance improves, while in others it worsens. Consequently, it is possible that the optimization algorithm does not improve the performance in certain DTs which is logical in this type of large-scale networks. This discrepancy is attributed to the fact that not all DTs are identical, since the position of aliens and STAs is unknown. This can be clearly seen in Fig. 13a where there are DTs where the average throughput per AP is generally no more than 20 Mbps and yet in other DTs built up to 40 Mbps. Therefore, it is imperative to follow this DT Instances strategy when there is no recent and updated data because the network can be very changeable and dynamic. In this way, many different possible scenarios are recreated along with different optimizations in order to arrive at the best possible channel allocation.

Once superior results are obtained, the 25 channel configurations that have been selected in each DT simulation are stored to learn which channels have manifested most frequently for each AP. Fig. 14 shows for each AP the channel that has been selected most often in the 25 DTs and the percentage of times it has come up. A higher percentage means that the selected channel is very likely to be the best choice, however if the percentage is low it means that there are several selectable channel options for that AP and it is more uncertain which is the best choice. More than 33% is considered a high percentage since an approach is being used where channels 1, 6 and 11 are selected.

Once the channels that appear most often in each AP are available, the simulation that most closely matches the channels that appear most often is searched for. As illustrated in Fig. 15 shows how in simulations 5, 6, 8, 13 and 25 there is a channel coincidence of more than 60% of the APs.

Among these simulations, the DT with the highest coincidence is sought only in the APs in which its percentage is greater than 50%, i.e., the simulation in which the uncertainty that the most appropriate channel is selected is the lowest possible in order to ensure as much as possible an improvement in the network. In this regard, simulation 25 stands out with more

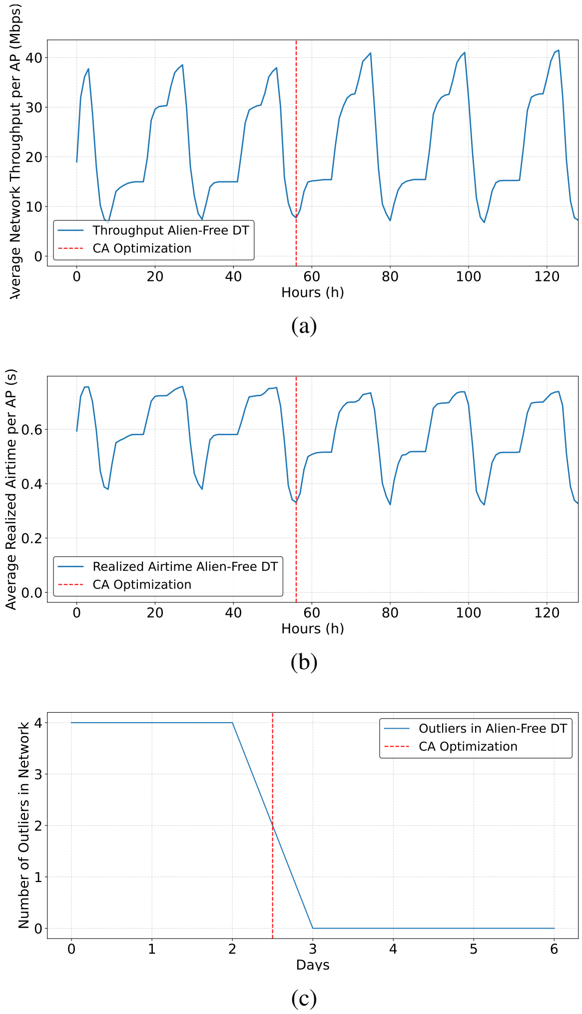
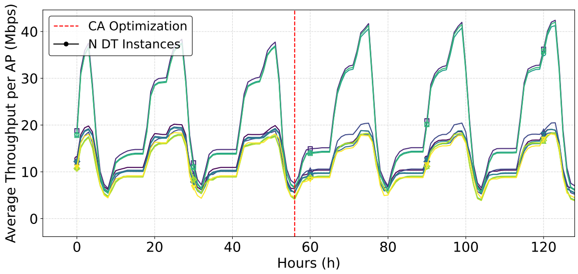
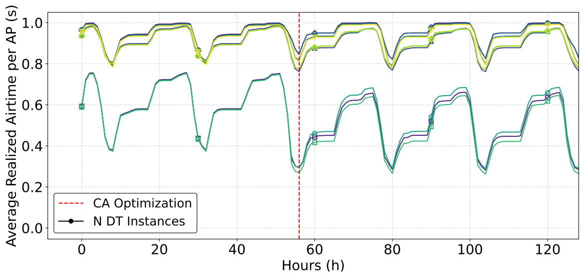


Fig. 12: Average performance with Alien-Free DT strategy: (a) Average throughput per AP; (b) Average airtime per AP; (c) Number of congestion outliers.



(a)



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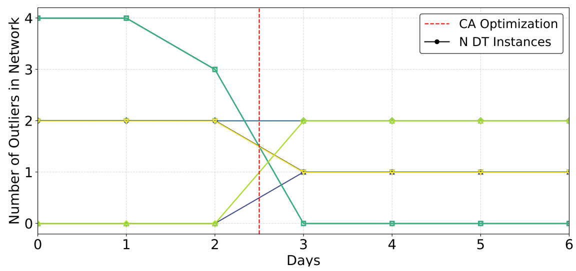

(c)

Fig. 13: Average performance with DT instances strategy: (a) Average throughput per AP; (b) Average airtime per AP; (c) Number of congestion outliers.

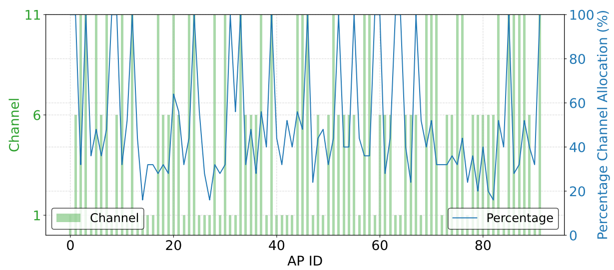

Fig. 14: Channels with the highest output for each AP in each simulation and its percentage.

than 80% of coincidence in these APs (see Fig. 16). Therefore, this combination has been chosen because it is the closest to the channels with the highest percentage of occurrence in the simulations.

With all this, the CA list used in that DT is obtained to upload it to the live operational network in an off-peak hour

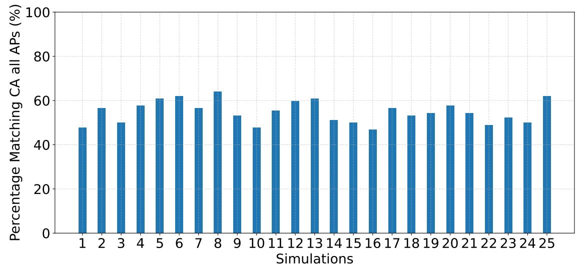

Fig. 15: Match of the simulations with the most frequently occurring channels.

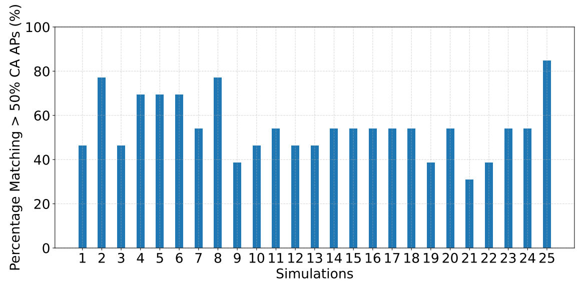

Fig. 16: Matching of simulations to the most frequent channels for APs with a greater than 50% match in the CA.

to monitor the network and see what results are obtained with this new configuration.

C. Live Operational Network Final Results

The network is monitored during the week preceding the new configuration upload. Once the new channel configuration obtained by MELODWIN is uploaded to the network, it is monitored for a week to observe any variations that the network may have undergone. After the test period, the network conditions are restored to the original conditions, and the network is also monitored during that period.

On a daily basis and taking the traffic global rush hour, the number of outliers exceeding the maximum noise level were recorded during the global rush hour. The lower the number of outliers, the better, as this indicates a smaller number of users at risk of underperformance.

As illustrated in Fig. 17, the count of outliers at the traffic rush hour for each weekday is presented on the leftmost side. It is evident that the implementation of the configuration derived from the MELODWIN system has resulted in a notable reduction in the number of noise outliers.

To confirm the effect, the week following the return to initial conditions was also recorded, as shown on the far right of 18. It can be seen how in the week when the MELODWIN system is applied, approximately 0.72 noise outliers are obtained on average compared to 1.28 and 0.88 in the week before and after the PoC respectively. In conclusion, the average number of noise outliers is only reduced during the PoC week when MELODWIN is used, with the number of noise outliers dropping by 18% to 44%.

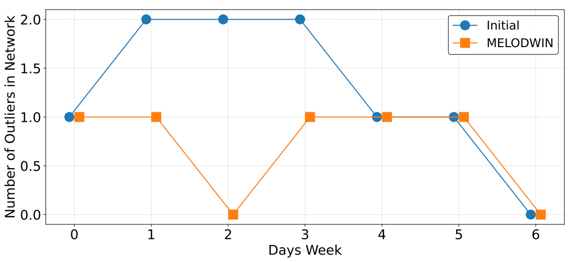

Fig. 17: Noise outliers results.

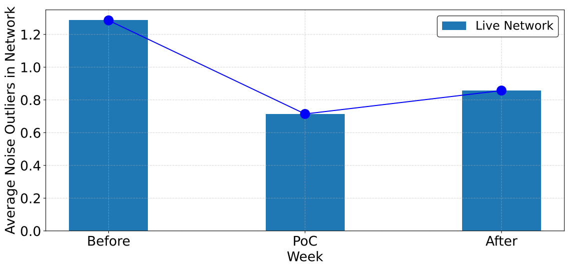

Fig. 18: Noise outliers results.

The second metric employed is the use or airtime accounted as the number of users that could use the network more intensively. Having a higher number of airtime outliers means that more users can make more intensive use of the resources, basically because more resources are available for them. This is also recorded on a weekday basis to decouple from the periodic human behavior.

As illustrated in Fig. 19, the majority of days exhibit a comparable or higher number of intensive users. It is important to acknowledge that this parameter is inherently influenced by human behavior, making it challenging to fully decouple from such factors.

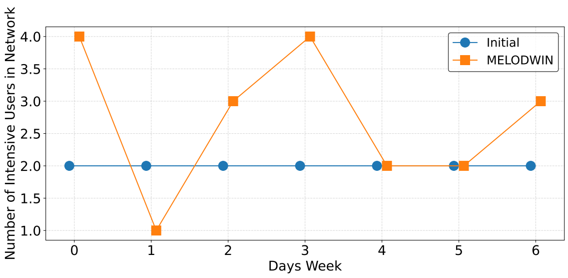

Fig. 19: Intensive outliers results.

Fig. 20 shows the average number of intensive users per week. It can be seen that clearly in the week in which MELODWIN is applied, it improves to an average of 2.8 compared to the previous week, which had 2 intensive users on average. The following week the change is also noticeable, since it drops to an average of 1.7. Therefore, an improvement of between 40% and 65% is obtained in intensive users.

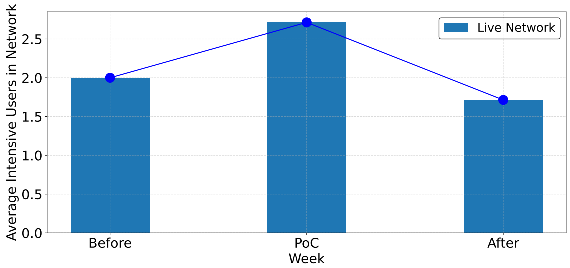

Fig. 20: Intensive outliers results.

The results presented indicate that by using MELODWIN on a live operational large-scale Wi-Fi network already optimized by an operator, noise outliers and heavy users have improved substantially. Although these two indicators measure different aspects of performance, their combined improvement shows a substantial improvement in overall network performance.

VI. Conclusions

This paper presented the MELODWIN system, an approach for CA optimization in large-scale Wi-Fi networks using RL supported by DT technology. The proposed solution leverages a dynamic loop between RL and DT, adapting strategies based on the age of data from the real network. This feature makes MELODWIN a versatile and robust framework for achieving optimal CA in live operational large-scale networks without degrading performance and with easy integration by adapting to the information coming from the network, while defining various rewards within the RL algorithm to guarantee network performance. The results demonstrated the practical effectiveness and potential real-world impact of the proposed optimization framework, showing improvements in CA and network performance through a comprehensive analysis and empirical testing. Furthermore, MELODWIN was validated not only at the simulation level against other approaches but was specially designed for live operational networks where it was validated with obvious improvement in noise and usage outliers in a network already optimized using current techniques, indicating a substantial impact of the proposed approach on the network.

Although this was a robust test case for large-scale networks, further validation on other live operational networks would have provided additional information and helped corroborate these results. Future work will explore additional mechanisms to enhance and extend the optimization process, including advanced techniques for structuring and abstracting network information, improved methods for creating accurate and scalable digital representations of real networks, and strategies for generating useful training data in data-scarce environments. These directions aim to build upon the current framework and support the development of more generalizable, data-efficient, and adaptable solutions for wireless network optimization.

Acknowledgment

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